Sensors & Systems — Lecture 8 Notes

Trilateration-based Sensors: BLE, Games-on-Track, GPS

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Trilateration

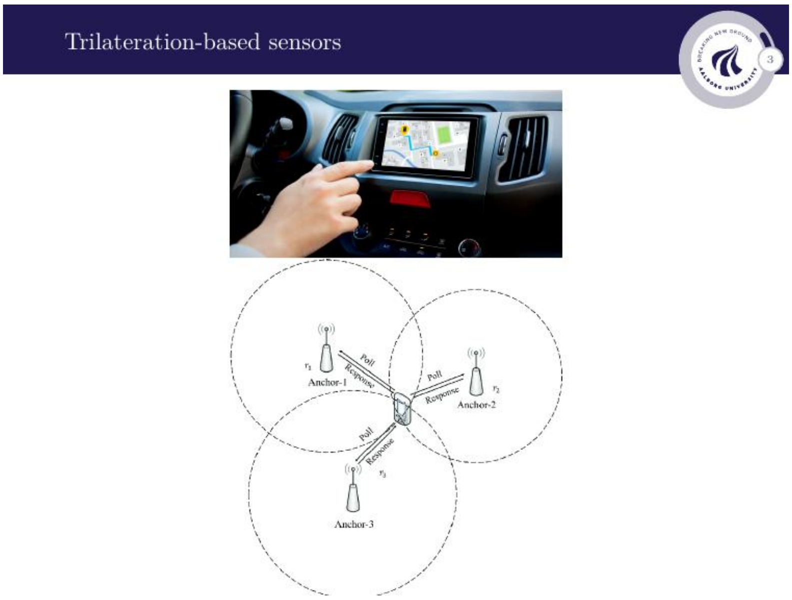

Figure 1: Three anchors at known positions each measure a range r_i to the unknown target via a Poll/Response exchange. Each range defines a circle; the intersection of all three gives the unique target location.

Trilateration estimates an unknown position from distance measurements to **anchors** at known coordinates. One range places the target on a circle (2D) or sphere (3D). A second narrows this to two candidate points. A **third** gives the unique solution. In 3D, four anchors are needed when an additional unknown exists (e.g. the receiver clock error in GPS).

Underwater Acoustic Positioning

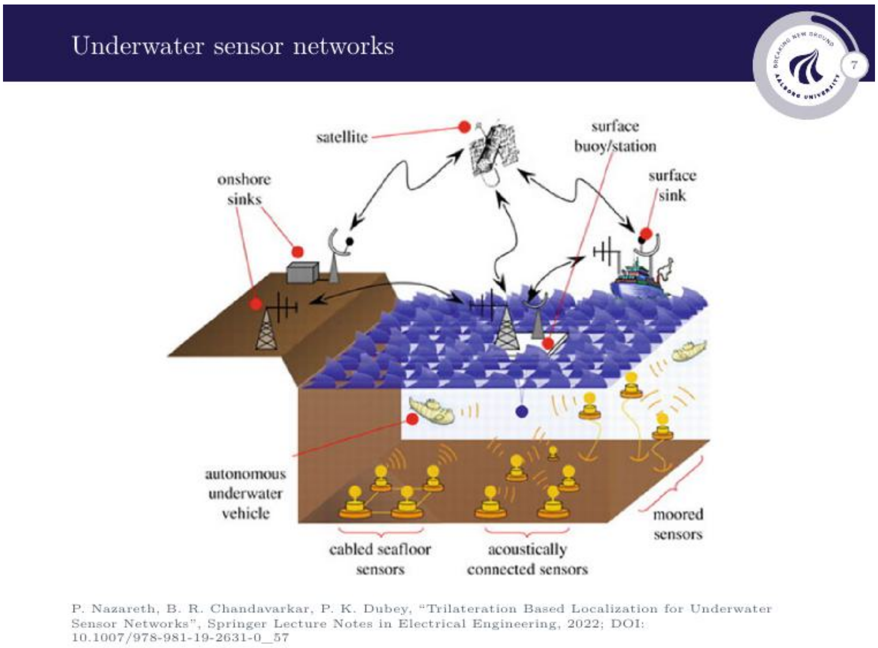

Figure 2: Underwater sensor network. Acoustic signals handle ranging between underwater nodes; radio relays the computed position to shore.

The same trilateration principle applies underwater, but radio is not an option — salt water absorbs radio waves within a few centimetres. **Acoustic signals** (sound) are used instead, with $c \approx 1500\,\mathrm{m/s}$, giving the same $r = c\,\Delta t/2$.

> **Two media, two roles — neither can replace the other. Acoustic** does the ranging underwater (sound travels hundreds of metres; radio absorbed within cm of salt water). **Radio** relays the already-computed position from a surface buoy to a satellite or shore station (sound too slow and short-range in air; radio travels at c over thousands of km). The surface buoy is the bridge — listening acoustically below, transmitting by radio above.

GPS — Coordinate Systems

ECEF and local coordinates

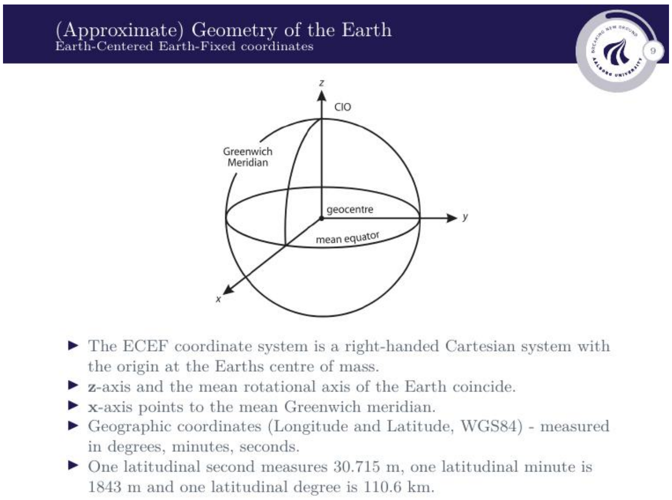

Figure 3: ECEF coordinate system. Origin at Earth's centre; z toward the north pole; x toward the Greenwich meridian. The frame rotates with the Earth so fixed surface points always have the same (x, y, z).

All GPS calculations use **ECEF** (Earth-Centred Earth-Fixed) — a Cartesian system rotating with the Earth. A point at longitude λ, latitude ϕ, radial distance γ from Earth's centre has ECEF coordinates:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \gamma \begin{bmatrix} \cos\phi\cos\lambda \\ \cos\phi\sin\lambda \\ \sin\phi \end{bmatrix}$$

where λ is the angle east/west of Greenwich ($-180°$ to $+180°$), ϕ is the angle north/south of the equator ($-90°$ to $+90°$), and $\gamma \approx R_E = 6{,}371\,\text{km}$ for points on the surface.

ECEF is good for the maths but hard to interpret in practice. **Local frames** centred on the observer make errors intuitive:

- **NED** (North-East-Down): standard in aviation; z points into the Earth so altitude increases as z decreases.
- **ENU** (East-North-Up): z points up, more natural for ground use.
- **AER** (Azimuth-Elevation-Range): describes a target by compass direction, elevation angle above horizon, and slant distance — used for pointing at satellites.

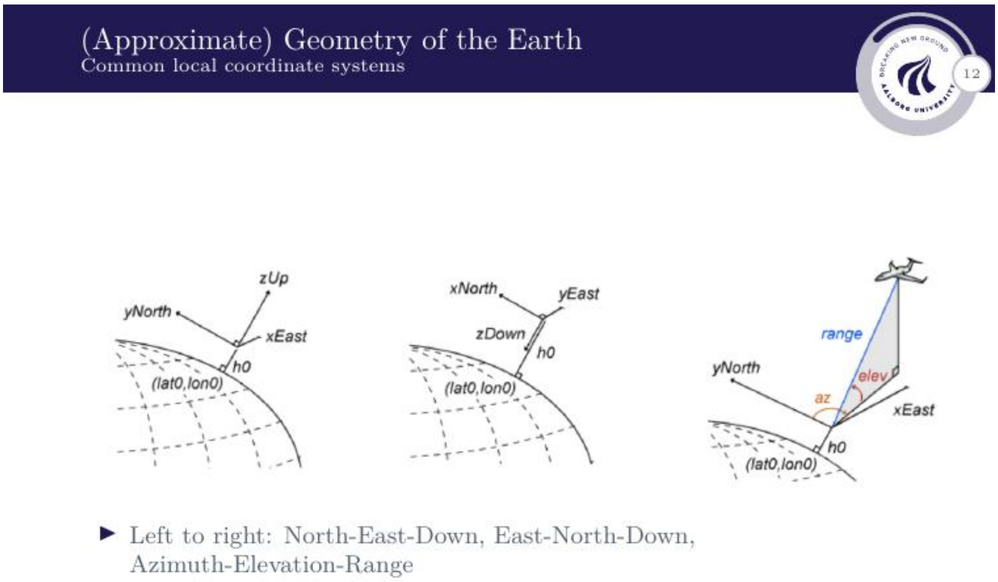

Figure 4: NED, ENU, and AER local coordinate systems, all centred at a reference point on the surface.

GPS — Pseudoranges and Why 4 Satellites

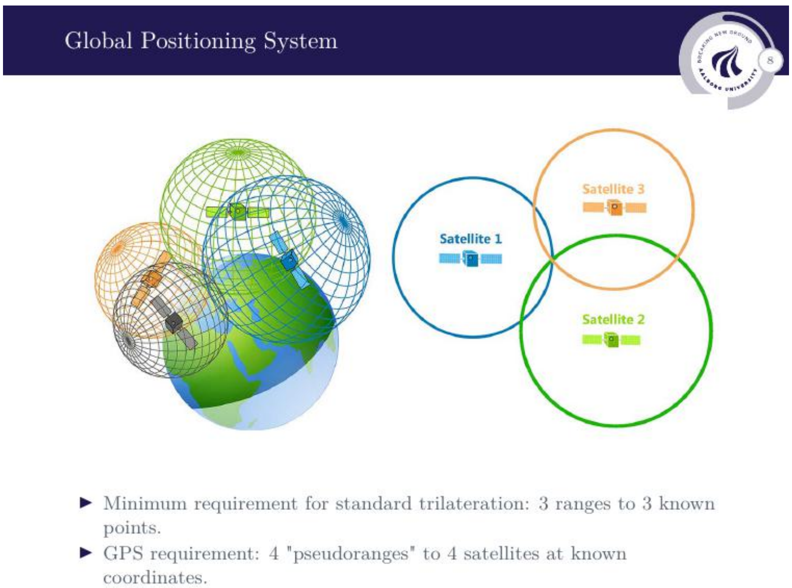

Figure 5: GPS trilateration in 3D. Three spheres give two candidate points; a fourth satellite resolves this and simultaneously solves for the unknown receiver clock error.

Standard 3D trilateration needs 3 ranges for 3 unknowns (x, y, z). GPS needs a **fourth satellite** because the receiver clock error is a fourth unknown. The satellite carries an **atomic clock** ($< 1\,\mu s$/day drift); the receiver uses a cheap **quartz oscillator** (tens of μs/second drift). Because the receiver clock is off by unknown τ, every range measurement is corrupted by the same $c\tau$, giving a **pseudorange**:

$$\boxed{P_S = \rho(t, t_S) + c\tau - c\tau_S}$$

Symbol	Name	Meaning
P_S	Pseudorange [m]	Measured apparent distance: raw $\Delta t \times c$
$\rho(t, t_S)$	True geometric range	$\sqrt{(x_S - x)^2 + (y_S - y)^2 + (z_S - z)^2}$
τ	Receiver clock error	Unknown offset of receiver quartz clock from GPS time
τ_S	Satellite clock error	Known — provided in Navigation Message as polynomial
t, t_S	True times	True GPS times of reception and transmission

Clock drift and the Navigation Message. Even the satellite's atomic clock drifts slightly. The GPS Control Segment (a global network of ground stations) tracks all satellites, measures their clock deviations, and uplinks correction coefficients packed into the Navigation Message as:

$$\delta t^s = a_0 + a_1(t - t_0) + a_2(t - t_0)^2$$

where a_0, a_1, a_2 are the offset, drift, and drift-rate coefficients, and t_0 is the reference epoch. The receiver reads and subtracts this correction, so satellite clock error is essentially a solved problem. The receiver clock error τ has no such correction and must be solved for alongside (x, y, z) — hence 4 satellites.

GPS — Signal and Cross-Correlation

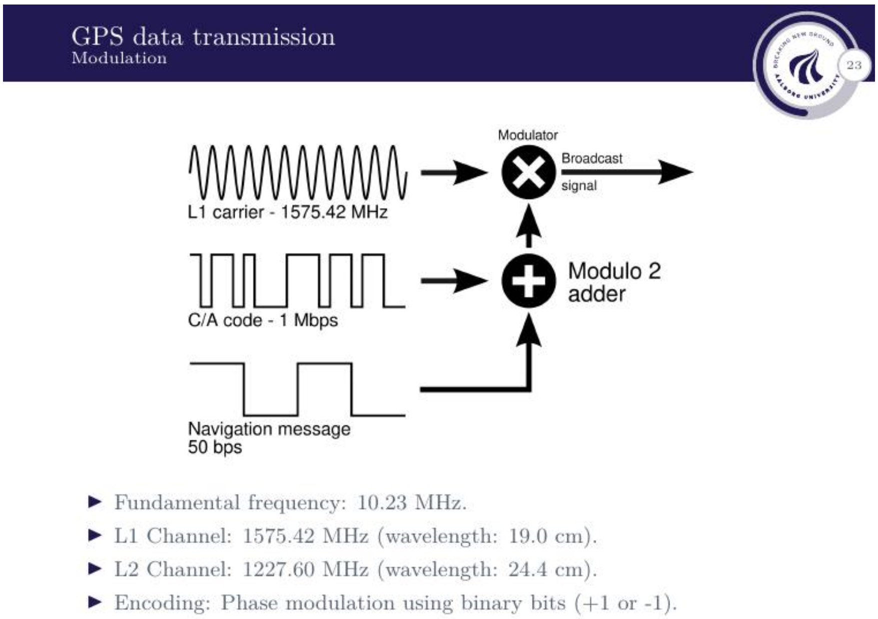

Figure 6: GPS signal structure. The L1 carrier (1575.42 MHz) is phase-modulated by the XOR of the C/A ranging code (1 Mbps) and the Navigation Message (50 bps).

The GPS signal is built in three layers, all derived from the atomic clock at $f_0 = 10.23\,\text{MHz}$:

- **L1 carrier** ($154 \times f_0 = 1575.42\,\text{MHz}$): the radio wave that carries everything through 20 000 km of space.
- **C/A code** ($f_0/10 = 1.023\,\text{Mbps}$): a 1023-chip pseudorandom sequence repeating every millisecond. Each satellite uses a *different* code (CDMA), allowing all satellites to share the same frequency without interfering. The fast chip rate gives microsecond-precision timing for ranging.
- **Navigation Message** (50 bps): the slow data payload carrying clock corrections, ephemeris, and almanac. Timing comes from the C/A code, not from these bits.

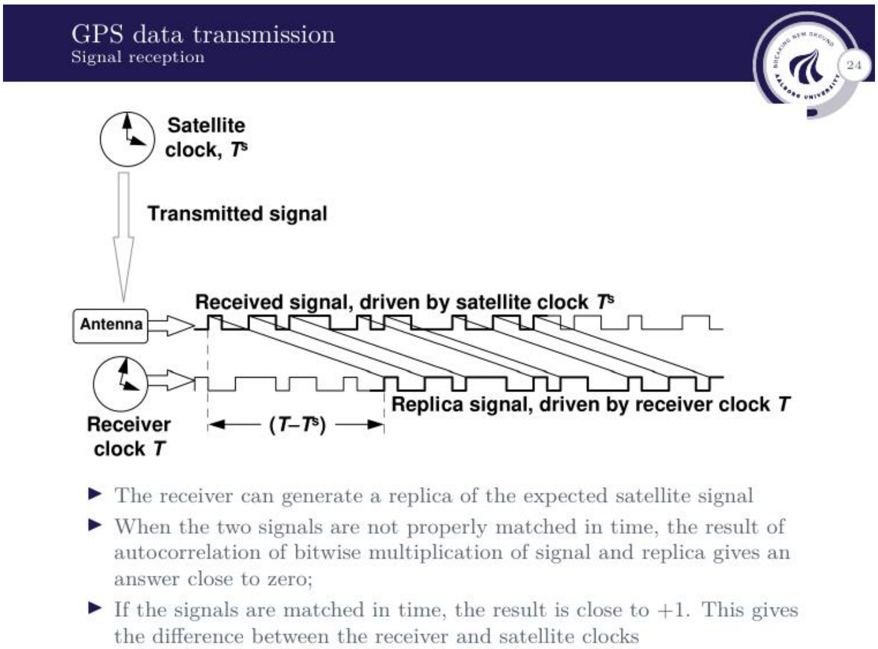

Figure 7: Signal reception via cross-correlation. The receiver generates a local replica of the satellite's PRN code and slides it in time. When misaligned the correlation is ≈ 0; when aligned it spikes to $+1$, giving the clock difference $T - T^s$ and hence the pseudorange.

The receiver measures pseudorange via **cross-correlation**:

$$R(k) = \frac{1}{N} \sum_{i=0}^{N-1} x_i \cdot y_{i-k}$$

where x_i is the received chip, y_{i-k} is the local replica delayed by k chips, and N is the number of chips summed. When the lag $k \neq \Delta t$ the pseudorandom sequences are misaligned and products nearly cancel ($R \approx 0$). When $k = \Delta t$ every chip multiplies its own copy giving $R = +1$ — a sharp spike. The lag at the spike is the travel time:

$$\Delta t = k \qquad \Rightarrow \qquad \rho = c\,\Delta t$$

This is exactly the same principle as Games-on-Track (chirp instead of PRN code): transmit a known pattern, generate a local copy, slide until they align — the required shift is the travel time.

GPS — Position Estimation and Error

Linearisation and least squares

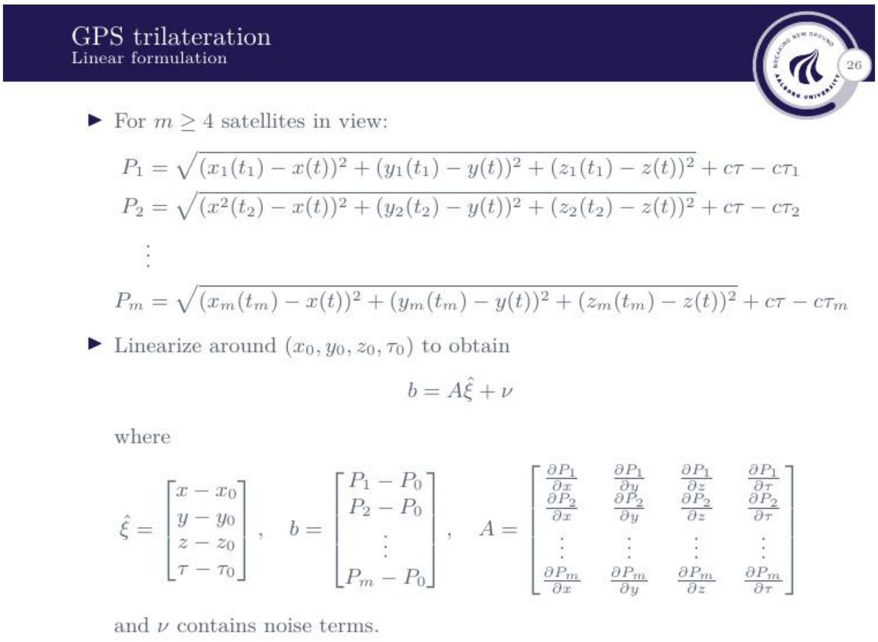

Figure 8: With $m \geq 4$ satellites, the pseudorange equations are linearised around an initial guess giving $b = A\hat{\xi} + \nu$.

With m satellites and satellite clock corrections applied, the pseudorange equations become:

$$\tilde{P}_k = \rho_k(x, y, z) + c\tau, \qquad k = 1, \dots, m$$

This is nonlinear in the unknowns (x, y, z, τ). Linearising around an initial guess (x_0, y_0, z_0, τ_0) and defining the correction vector $\hat{\xi}$ and measurement residual b:

$$\hat{\xi} = \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \\ \tau - \tau_0 \end{bmatrix}, \qquad b = \begin{bmatrix} \tilde{P}_1 - P_{0,1} \\ \vdots \\ \tilde{P}_m - P_{0,m} \end{bmatrix}$$

gives the linear system $b = A\hat{\xi} + \nu$, where ν is the fit residual. Each row of the Jacobian A is a unit vector pointing from the receiver toward satellite k, with c appended for the clock column:

$$A_k = \begin{bmatrix} -\dfrac{x_S^{(k)} - x_0}{\rho_{0,k}} & -\dfrac{y_S^{(k)} - y_0}{\rho_{0,k}} & -\dfrac{z_S^{(k)} - z_0}{\rho_{0,k}} & c \end{bmatrix}$$

where $\rho_{0,k}$ is the predicted range to satellite k from the initial guess, and $(x_S^{(k)}, y_S^{(k)}, z_S^{(k)})$ is the known ECEF position of satellite k. With $m > 4$ the system is overdetermined; minimising $\|\nu\|^2$ gives the **least-squares solution**:

$$\boxed{\hat{\xi} = (A^\top A)^{-1} A^\top b}$$

Why more satellites help and why geometry matters (DOP). With exactly 4 satellites any noise goes straight into the position estimate. With $m > 4$, extra measurements are redundant but useful — least squares averages out random noise across all satellites. The conditioning of $A^\top A$ quantifies satellite geometry: good spread across the sky $\to$ $A^\top A$ well-conditioned $\to$ small position errors. Poor spread $\to$ $A^\top A$ nearly singular $\to$ large errors. This is exactly what **DOP** (Dilution of Precision) measures — a high DOP means bad geometry and amplified errors.

Error propagation and covariance

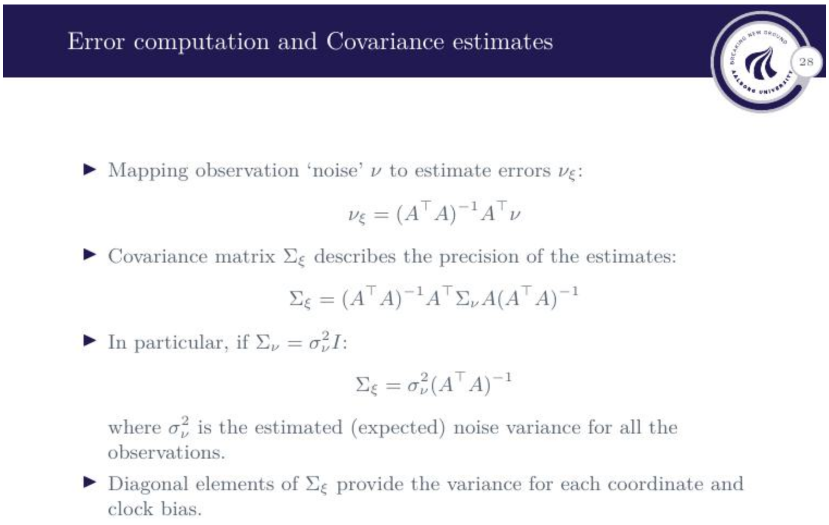

Figure 9: Pseudorange measurement noise Σ_ν propagated through the least-squares formula gives the position covariance Σ_ξ.

The estimation error propagates from pseudorange noise ν as:

$$\nu_\xi = (A^\top A)^{-1} A^\top \nu$$

Taking the expectation of the outer product gives the **covariance matrix** of position errors:

$$\Sigma_\xi = (A^\top A)^{-1} A^\top \Sigma_\nu A \, (A^\top A)^{-1}$$

where Σ_ν is the $m \times m$ covariance of the pseudorange noise. Assuming equal, independent noise on all satellites ($\Sigma_\nu = \sigma_\nu^2 I$) this simplifies to:

$$\Sigma_\xi = \sigma_\nu^2 \, (A^\top A)^{-1}$$

The **diagonal elements** give individual variances: $\sigma_x^2, \sigma_y^2, \sigma_z^2$ for position and σ_τ^2 for the clock error estimate. The shape of the uncertainty ellipsoid is determined entirely by satellite geometry $(A^\top A)^{-1}$; noise level σ_ν^2 scales the whole thing uniformly.

Uncertainty ellipsoid in local coordinates

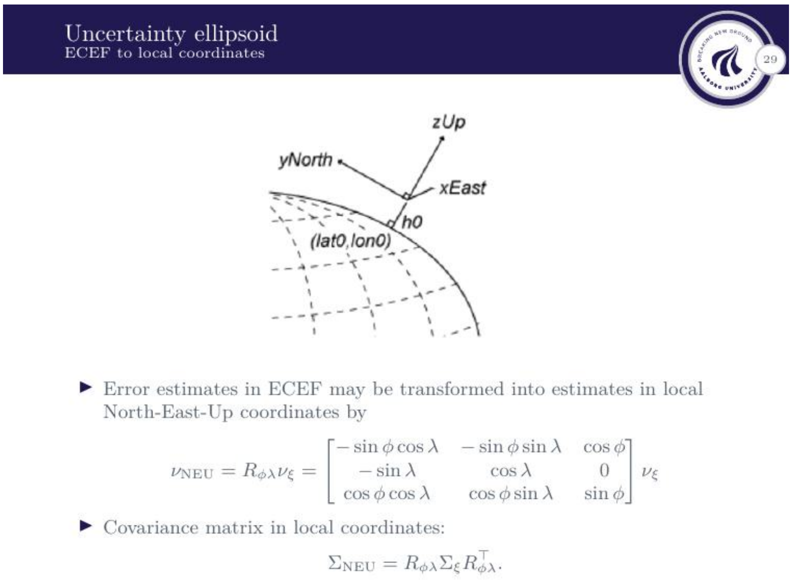

Figure 10: The ECEF covariance Σ_ξ is rotated into local North-East-Up coordinates using $R_{\phi\lambda}$ for a more intuitive decomposition into horizontal and vertical errors.

An error of 5 m in the ECEF x-direction is hard to interpret. Rotating into local North-East-Up (NEU) coordinates makes it meaningful:

$$\nu_{\mathrm{NEU}} = R_{\phi\lambda}\,\nu_\xi, \qquad \Sigma_{\mathrm{NEU}} = R_{\phi\lambda}\,\Sigma_\xi\,R_{\phi\lambda}^\top$$

where the rotation matrix at latitude ϕ and longitude λ is:

$$R_{\phi\lambda} = \begin{bmatrix} -\sin\phi\cos\lambda & -\sin\phi\sin\lambda & \cos\phi \\ -\sin\lambda & \cos\lambda & 0 \\ \cos\phi\cos\lambda & \cos\phi\sin\lambda & \sin\phi \end{bmatrix}$$

The diagonal of Σ_{NEU} now gives $\sigma^2_{\mathrm{North}}$, σ^2_{East}, σ^2_{Up} — directly telling you the horizontal spread and vertical offset of the uncertainty ellipsoid.

Tracking moving targets with a Kalman filter

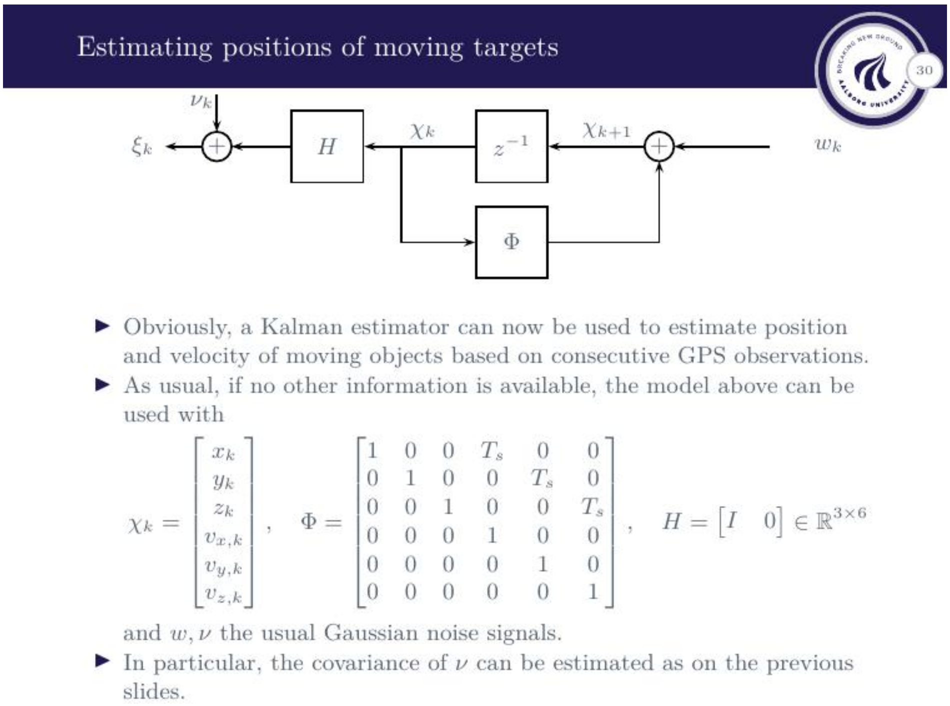

Figure 11: A Kalman filter takes consecutive GPS position estimates and infers both position and velocity using a constant-velocity motion model.

GPS gives a new position estimate every second; a **Kalman filter** turns this into a smooth position-and-velocity track. With no other information, a constant-velocity model is used:

$$\chi_k = \begin{bmatrix} x_k \\ y_k \\ z_k \\ v_{x,k} \\ v_{y,k} \\ v_{z,k} \end{bmatrix}, \quad \Phi = \begin{bmatrix} 1 & 0 & 0 & T_s & 0 & 0 \\ 0 & 1 & 0 & 0 & T_s & 0 \\ 0 & 0 & 1 & 0 & 0 & T_s \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} I & 0 \end{bmatrix} \in \mathbb{R}^{3\times 6}$$

where χ_k is the state vector of position and velocity, Φ is the state transition matrix with sample time T_s, and H extracts position only (GPS measures position, not velocity directly — the filter infers velocity from how position changes between steps). The covariance Σ_ν from the previous section is used directly as the measurement noise covariance in the Kalman filter.

Lecture 8 — Key Takeaways

- **BLE RSSI**: range via $\hat{\rho} = 10^{(A-S)/(10\gamma)}$. No timing hardware; 1–5 m accuracy indoors due to multipath and γ uncertainty.
- **Games-on-Track / GPS**: TOF via cross-correlation of a known pattern (chirp / PRN code). Locks onto first-arriving signal; robust to multipath; needs dedicated timing hardware.
- **Underwater**: radio replaced by acoustic ($c \approx 1500\,\mathrm{m/s}$); same TOF principle.

Radio used only at surface to relay position to shore.

- **GPS pseudorange**: $P_S = \rho + c\tau - c\tau_S$. Satellite clock error τ_S corrected via Navigation Message polynomial. Receiver clock error τ solved alongside (x, y, z) — requires 4 satellites minimum.
- **Least squares**: $\hat{\xi} = (A^\top A)^{-1} A^\top b$, where each row of A is a unit vector toward a satellite with c appended. More satellites reduce noise; satellite geometry (DOP) determines how much noise gets amplified into position error.
- **Covariance**: $\Sigma_\xi = \sigma_\nu^2 (A^\top A)^{-1}$; rotated to local NEU for interpretable horizontal/vertical errors. Fed directly into a Kalman filter for tracking moving targets.
- **ECEF** rotates with Earth; local frames (NED, ENU, AER) express errors relative to the observer.